

Design of experiment for model discrimination: SFE case, part II

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ABSTRACT

This study investigates the process of chamomile oil extraction from flowers. A parameter-distributed model consisting of a set of partial differential equations is used to describe the governing mass transfer phenomena in a solid-fluid environment under supercritical conditions using carbon dioxide as a solvent. The concept of quasi-one-dimensional flow is applied to reduce the number of spatial dimensions. The flow is assumed to be uniform across any cross-section, although the area available for the fluid phase can vary along the extractor. The physical properties of the solvent are estimated from the Peng-Robinson equation of state. Based on the set of laboratory experiments performed under multiple constant operating conditions: 30 – 40 °C, 100 – 200 bar, and $3.33 - 6.67 \cdot 10^{-5}$ kg/s, two process models are developed: a first-principle and a data-driven model. The goal of this work is to design a dynamic experiment to discriminate between two competing process models.

1. Introduction

Supercritical CO₂ is defined as carbon dioxide that is pressurized and heated above its critical point (31.1 °C, 74 bar). Depending on the operating conditions, the fluid properties such as viscosity and density can vary, which leads to multiple industrial applications of CO₂.

One of the most popular applications of supercritical CO₂ is the extraction of essential oils, as described by many researchers, for example, by Sodeifian and Sajadian [1], Reverchon et al. [2] or Sovova [3]. Traditional methods, such as distillation and organic solvent extraction, are commonly employed but have drawbacks. Distillation, involves high temperatures that can lead to the thermal degradation of heat-sensitive compounds. This limitation has increased the popularity of alternative techniques, such as supercritical fluid extraction. Supercritical CO₂ is appealing due to its distinctive properties: it is inflammable, non-toxic and non-corrosive. Supercritical fluids can exhibit both gas- and liquid-like properties, allowing for adjustable dissolving power through changes in operating conditions.

The applications of supercritical carbon dioxide are not limited only to an extraction process but can also be used for impregnation as described by Weidner [4], Machado et al. [5] or Fathi et al. [6]. Impregnation is defined as modifying the properties of bulk substances by physically or chemically binding/adsorbing impregnates to a bulk material or surface, such as the hydrophobization of surfaces. The main advantage of using supercritical CO₂ is that after depressurization, it desorbs from the surface and evaporates, leaving a solvent-free product. On the other hand, the main disadvantage of using carbon dioxide for impregnation is the low solubility of many drugs of interest.

Another application of supercritical CO₂ is nanoparticles formation as investigated by Padrela et al. [7], Franco and De Marco [8], Saadati Ardestani et al. [9] or Sodeifian

et al. [10]. Supercritical carbon-dioxide-assisted technologies enable the production of different morphologies of different sizes, including nanoparticles and nanocrystals, by modulating operating conditions. Supercritical fluid-based processes have advantages over techniques conventionally employed to produce nanosized particles or crystals, such as reduced use of toxic solvents. Moreover, the CO₂ is completely removed from the final product by simple depressurization.

This study investigates the extraction of essential oil from chamomile flowers (*Matricaria chamomilla* L.) via supercritical fluid extraction techniques and the modelling of this process. Chamomile is a medicinal herb widely cultivated in southern and eastern Europe — in countries such as Germany, Hungary, France and Russia. It can be found outside Europe, for instance in Brazil as discussed by Singh et al. [11]. This plant is distinguished by its hollow, bright gold cones, housing disc or tubular florets and surrounded by about fifteen white ray or ligulate florets. Chamomile has been used for its medicinal benefits, serving as an anti-inflammatory, antioxidant, mild astringent, and healing remedy. Extracts of chamomile are widely used to calm nerves and mitigate anxiety, hysteria, nightmares, insomnia and other sleep-related conditions, according to Srivastava [12]. Orav et al. [13] reported that oil yields from dried chamomile samples ranged from 0.7 to 6.7 mL/kg. The highest yields of essential oil, between 6.1 and 6.7 mL/kg, were derived from chamomile sourced from Latvia and Ukraine. In comparison, chamomile from Armenia exhibited a lower oil content of 0.7 mL/kg.

The literature offers various mathematical models to describe the extraction of valuable compounds from biomass. Selecting a process model is case-to-case dependent and requires analysis of each model's specific assumptions about mass transfer and thermodynamic equilibrium.

Goto et al. [14] presented the Shrinking Core (SC) model, which describes a process of irreversible desorption that is followed by diffusion through the pores of a porous solid. When the mass transfer rate of the solute in the non-extracted inner region is significantly slower than in the outer

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region, where most of the solute has already been extracted, or when the solute concentration exceeds its solubility in the solvent, a distinct boundary may form between the inner and outer regions. As extraction progresses, the core of the inner region shrinks. The model envisions supercritical CO_2 extraction as a sharp, inward-moving front, with a completely non-extracted core ahead of the front and a fully extracted shell behind it.

Sovova [3] proposed The Broken-and-Intact Cell (BIC) model, which assumes that a portion of the solute, initially stored within plant structures and protected by cell walls, is released during the mechanical breakdown of the material. The solute located in the region of broken cells near the particle surface is directly exposed to the solvent, while the core of the particle contains intact cells with undamaged walls. This model describes three extraction phases: a fast extraction phase for accessible oil, a transient phase, and a slow phase controlled by diffusion. The model has been successfully applied to the extraction of grape oil (Sovová et al. [15]) and caraway oil (Sovova et al. [16]).

The Supercritical Fluid Extraction (SFE) process can be treated similarly to heat transfer, considering solid particles like hot balls cooling down in a uniform environment. Bartle et al. [17] introduced the hot ball diffusion (HBD) model, where spherical particles with uniformly distributed solute diffuse similarly to heat diffusion. Unlike the BIC model, where solute is readily available on the particle surface, the HBD model is suited for systems with small quantities of extractable materials and is not limited by solubility. The model is particularly relevant when internal diffusion controls mass transfer, allowing results from single particles to be extended to the entire bed under uniform conditions. Reverchon et al. [2] have further elaborated on the HBD model and used it to simulate extraction processes for natural materials.

Reverchon [18] proposed a model for extraction of essential oils, which are mainly located inside the vegetable cells in organules called vacuoles. Only a small fraction of essential oil might be near the particle surface due to the breaking up of cells during grinding or in epidermal hairs located on the leaf surface. The fraction of oil freely available on the particle surface should not be significant in the case of SFE from leaves. Consequently, SFE of essential oil from leaves should be mainly controlled by the internal mass-transfer resistance. Therefore, the external mass-transfer coefficient was neglected in the development of the model of Reverchon [18]. The mass balances were developed in the additional hypotheses that the axial dispersion can be neglected and that the solvent density and flow rate are constant along the bed.

This work builds upon the linear kinetic model suggested by Reverchon [18], deriving fundamental governing equations to develop a comprehensive model for the chamomile oil extraction process. This model aims for control-oriented simplicity, assuming a semi-continuous operation within a cylindrical vessel. The process involves a supercritical solvent being pumped through a fixed bed

of finely chopped biomass to extract the solute, followed by separation of the solvent and solute in a flush drum to collect the extract. Parameters such as pressure (P), feed flow rate (F) and inlet temperature (T_{in}) are adjustable and measurable, while the outlet temperature (T_{out}) and the amount of product at the outlet can only be monitored. Figure 1 presents a simplified process flow diagram.

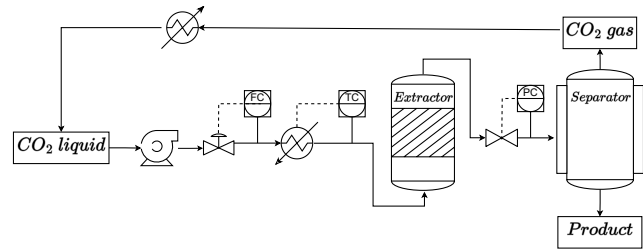


Figure 1: Process flow diagram.

In the study of dynamical systems, selecting the most accurate model to represent a physical process is crucial for effective prediction, control, and understanding of underlying mechanisms. When multiple models are trained on the same dataset but exhibit subtle structural differences, their responses may diverge in nuanced ways. Traditional model selection criteria often fail to capture these subtleties, making it necessary to employ a robust statistical framework for optimal model discrimination.

The process of discriminating between competing models typically follows a sequential approach, involving two iterative steps. The first is the design phase, where a discrimination criterion is used to select the next experiment. This criterion, which reflects the information content of potential experimental conditions, helps identify the experiment that is expected to provide the most information for distinguishing between models. Once the experiment is conducted, the second step—data analysis—is performed. A stopping rule is then applied to determine if one model can be deemed significantly better than the others. If no clear distinction is made, the process is repeated. Most model discrimination criteria rely on either maximum divergence or maximum entropy principles.

Building on information theory and a Bayesian approach, Box and Hill [19] introduced a sequential design procedure for model discrimination. Since entropy measures the amount of available information about a system, additional experiments should be selected to maximize the expected change in entropy, thereby increasing the information obtained about the system through successive experiments.

Buzzi-Ferraris and Forzatti [20] observed that the performance of Box's sequential design for model discrimination depends on the order in which experimental observations are processed. This is due to the recursive nature of updating model probabilities. As model probabilities should be independent of the data presentation order, the authors noted that experiments should be selected in regions where differences in prediction variances are large, rather than where differences in model responses are most pronounced.

They applied this approach to two cases: ammonia synthesis from nitrogen and hydrogen, and vapor-phase chlorination of tetrachloroethane using activated silica gel at 200 °C.

Chen and Asprey [21] proposed a novel method for designing optimally informative dynamic experiments aimed at discriminating among several competing multiresponse nonlinear dynamic models, which are generally described by systems of differential and algebraic equations. Their approach extends Buzzi-Ferraris and Forzatti [20]' discrimination criterion to dynamic systems, reformulating the experimental design problem as an optimal control problem. This method was applied to the fed-batch fermentation of Baker's yeast, with the goal of discriminating between four competing models by identifying optimal profiles for the dilution factor and inlet substrate concentration.

2. Materials and methods

2.1. Governing equations

Following the work of Anderson [22], the governing equations for a quasi-one-dimensional flow were derived. A quasi-one-dimensional flow refers to a fluid flow scenario assuming that the flow properties are uniformly distributed across any cross-section. This simplification is typically applied when the flow channel's cross-sectional area changes, such as through irregular shapes or partial filling of an extractor. According to this assumption, velocity and other flow properties change solely in the flow direction.

As discussed by Anderson [23], all flows are compressible, but some of them can be treated as incompressible since the velocities are low. This assumption leads to the incompressible condition: $\nabla \cdot u = 0$, which is valid for constant density (strict incompressible) or varying density flow. The assumption allows for removing acoustic waves and large perturbations in density and/or temperature. In the 1-D case, the incompressibility condition becomes $\frac{du}{dz} = 0$, so the fluid velocity is constant along the z -direction.

The set of quasi-one-dimensional governing equations in Cartesian coordinates is described by Equations 1 - 3:

$$\frac{\partial(\rho_f A_f)}{\partial t} + \frac{\partial(\rho_f A_f v)}{\partial z} = 0 \quad (1)$$

$$\frac{\partial(\rho_f v A_f)}{\partial t} + \frac{\partial(\rho_f A_f v^2)}{\partial z} = -A_f \frac{\partial P}{\partial z} \quad (2)$$

$$\frac{\partial(\rho_f e A_f)}{\partial t} + \frac{\partial(\rho_f A_f v e)}{\partial z} = -P \frac{\partial(A_f v)}{\partial z} + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) \quad (3)$$

where ρ_f is the density of the fluid, A_f is the function which describes a change in the cross-section, v is the velocity, P is the total pressure, e is the internal energy of the fluid, t is time and z is the spatial direction.

2.2. Extraction model

2.2.1. Continuity equation

The previously derived quasi-one-dimensional continuity equation (Equation 1) is redefined by incorporating the function $A_f = A\phi$. This modification distinguishes constant and varying terms, where the varying term accounts

for changes in the cross-sectional area available for the fluid. Equation 4 shows the modified continuity equation:

$$\frac{\partial(\rho_f \phi)}{\partial t} + \frac{\partial(\rho_f v A \phi)}{\partial z} = 0 \quad (4)$$

where A is the total cross-section of the extractor and ϕ describes porosity along the extractor.

Assuming that the mass flow rate is constant in time, the temporal derivative becomes the mass flux F , and the spatial derivative can be integrated along z as

$$\int \frac{\partial(\rho_f v A \phi)}{\partial z} dz = F \rightarrow F = \rho_f v A \phi \quad (5)$$

To simplify the system dynamics, it is assumed that F is a control variable and affects the whole system instantaneously (due to $\nabla \cdot u = 0$), which allows finding the velocity profile that satisfies mass continuity based on F , ϕ and ρ_f :

$$v = \frac{F}{\rho_f A \phi} \quad (6)$$

Similarly, superficial velocity may be introduced:

$$u = v\phi = \frac{F}{\rho_f A} \quad (7)$$

The fluid density ρ_f can be obtained from the Peng-Robinson equation of state if the temperature and thermodynamic pressure are known along z . Variation in fluid density may occur due to pressure or inlet temperature changes. In a non-isothermal case, in Equations 6 and 7 ρ_f is considered the average fluid density along the extraction column.

2.2.2. Mass balance for the fluid phase

Equation 8 describes the movement of the solute in the system, which is constrained to the axial direction due to the quasi-one-dimensional assumption. Given that the solute concentration in the solvent is negligible, the fluid phase is described as pseudo-homogeneous, with properties identical to those of the solvent itself. It is also assumed that the thermodynamic pressure remains constant throughout the device. The analysis further simplifies the flow dynamics by disregarding the boundary layer near the extractor's inner wall. This leads to a uniform velocity profile across any cross-section perpendicular to the axial direction. Thus, the mass balance equation includes convection, diffusion and kinetic terms representing the fluid phase behaviour:

$$\frac{\partial c_f}{\partial t} + \frac{1}{\phi} \frac{\partial(c_f u)}{\partial z} = \frac{1 - \phi}{\phi} r_e + \frac{1}{\phi} \frac{\partial}{\partial z} \left(D_e^M \frac{\partial c_f}{\partial z} \right) \quad (8)$$

where c_f represents the solute concentration in the fluid phase, r_e is the mass transfer kinetic term and D_e^M is the axial diffusion coefficient.

2.2.3. Mass balance for the solid phase

As given by Equation 9, the solid phase is considered stationary, without convection and diffusion terms in the mass balance equation. Therefore, the only significant term in this equation is the kinetic term of Equation 10, which connects the solid and fluid phases. For simplicity, the extract is represented by a single pseudo-component:

$$\frac{\partial c_s}{\partial t} = \underbrace{r_e}_{\text{Kinetics}} \quad (9)$$

2.2.4. Kinetic term

As the solvent flows through the fixed bed, CO₂ molecules diffuse into the pores, adsorb on the inner surface and form a film due to solvent-solid matrix interactions. The dissolved solute diffuses from the particle's core through the solid-fluid interface, the pore and the film into the bulk. Figure 2 shows the mass transfer mechanism, where the mean solute concentration in the solid phase is denoted as c_s , and the equilibrium concentrations at the solid-fluid interface are denoted as c_s^* and c_p^* for the solid and fluid phases, respectively. The concentration of the solutes in the fluid phase in the centre of the pore is denoted as c_p . As the solute diffuses through the pore, its concentration changes, reaching c_{pf} at the opening. Then, the solute diffuses through the film around the particle and reaches bulk concentration c_f . The two-film theory describes the solid-fluid interface inside the pore. The overall mass transfer coefficient can be determined from the relationship between the solute concentration in one phase and its equilibrium concentration.

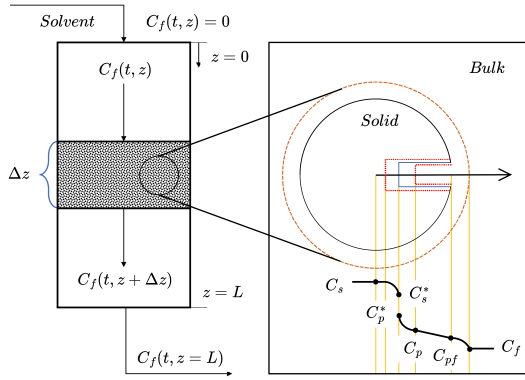


Figure 2: Mass transfer mechanism.

Bulley et al. [24] suggest a process where the driving force for extraction is given by the difference between the concentration of the solute in the bulk, c_f , and in the centre of the pore, c_p^* . The concentration c_p^* is in equilibrium with c_s according to the equilibrium relationship. The rate of extraction is thus $r_e \left(c_f - c_p^*(c_s) \right)$. In contrast, Reverchon [18] proposes a driving force given by the difference between c_s and c_p^* . As given by Equation 10, the concentration c_p^* is determined by the equilibrium relationship with c_f :

$$r_e = \frac{D_i}{\mu l^2} (c_s - c_p^*) \quad (10)$$

where μ is sphericity, l a characteristic dimension of particles that can be defined as $l = r/3$, r is the mean particle radius, ρ_s is the solid density, D_i corresponds to the overall diffusion coefficient and c_p^* is the concentration at the solid-fluid interface (which according to the internal resistance model is supposed to be at equilibrium with the fluid phase).

According to Bulley et al. [24], a linear equilibrium relationship (Equation 11) can be used to find the equilibrium concentration of the solute in the fluid phase c_f^* based on the concentration of the solute in the solid phase c_s :

$$c_f^* = k_p c_s \quad (11)$$

The volumetric partition coefficient k_p acts as an equilibrium constant between the solute concentration in one phase and the corresponding equilibrium concentration at the solid-fluid interphase. Spiro and Kandiah [25] proposed to define the mass partition coefficient k_m as

$$k_m = \frac{k_p \rho_s}{\rho_f} \quad (12)$$

According to Reverchon [18], the kinetic term becomes

$$r_e = -\frac{D_i}{\mu l^2} \left(c_s - \frac{\rho_s c_f}{k_m \rho_f} \right) \quad (13)$$

2.2.5. Uneven solute's distribution in the solid phase

Following the idea of the Broken-and-Intact Cell (BIC) model (Sovova [26]), the internal diffusion coefficient D_i is considered to be a product of the reference value of D_i^R and the exponential decay function γ , as given by Equation 14:

$$D_i = D_i^R \gamma(c_s) = D_i^R \exp \left(\Upsilon \left(1 - \frac{c_s}{c_{s0}} \right) \right) \quad (14)$$

where Υ describes the curvature of the decay function. Equation 15 describes the final form of the kinetic term:

$$r_e = -\frac{D_i^R \gamma}{\mu l^2} \left(c_s - \frac{\rho_s c_f}{k_m \rho_f} \right) \quad (15)$$

The γ function limits the solute's availability in the solid phase. Similarly to the BIC model, the solute is assumed to be contained in the cells, some of which are open because the cell walls were broken by grinding, with the rest remaining intact. The diffusion of the solute from a particle's core takes more time than the diffusion of the solute close to the outer surface. The same idea can be represented by the decaying internal diffusion coefficient, where the decreasing term is a function of the solute concentration in the solid.

An alternative interpretation of the decay function γ involves considering the porous structure of the solid particles, where the pores are initially saturated with the solute. During extraction, the solute within these pores gradually dissolves into the surrounding fluid. Initially, the solute molecules near the pore openings dissolve and diffuse rapidly due to the short diffusion paths. As the extraction progresses, the dissolution front moves deeper into the pore structure, and solute from the inner regions of the pores begins to dissolve. The diffusion of solute molecules from the interior of the pores to the external fluid becomes progressively slower because the effective diffusion path length increases. This lengthening of the diffusion path enhances the mass transfer resistance, reducing the overall diffusion rate.

In an extreme case, this model could be compared with the Shrinking Core Model presented by Goto et al. [14], where the particle radius decreases as the solute content in the solid phase diminishes. In the SC model, the reduction in particle size leads to significant changes in both the diffusion path length and the surface area available for mass transfer. The diminishing particle size increases the diffusion path within the remaining solid core and decreases the external

surface area, both of which contribute to a slower extraction rate. By comparing this to the varying diffusion coefficient in our model, some conceptual similarities can be noticed.

2.2.6. Empirical correlations

The empirical correlations for D_i and Υ were derived by Sliczniuk and Oinas [27] and validated for temperatures between 30 – 40 °C, pressures between 100 – 200 bar, and mass flow rates between $3.33 - 6.67 \cdot 10^{-5}$ kg/s. Figures 3 and 4 show the results of multiple linear regression applied to solutions of parameter estimation and selected independent variables. The region marked with the white dashed line represents the confidence region, where the model has been tested. Both correlations should be equal of greater than zero, to avoid unphysical behaviour such as the reverse mass transfer. The multiple linear regression functions are combined with the rectifier function to ensure the non-negativity.

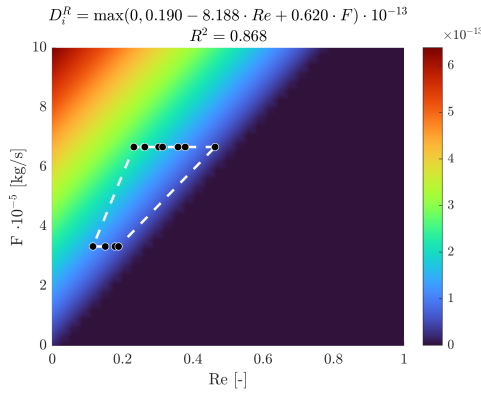


Figure 3: Multiple linear regression $D_i^R = f(Re, F)$

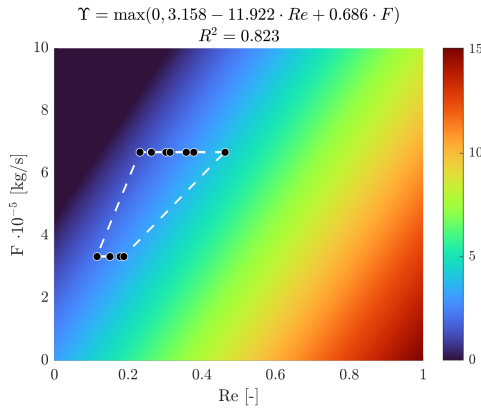


Figure 4: Multiple linear regression $\Upsilon = f(Re, F)$

2.2.7. Heat balance

The heat balance equation describe the evolution of the enthalpy in the system and it is given by Equation 16

$$\frac{\partial (\rho_f h A_f)}{\partial t} = - \frac{\partial (\rho_f h A_f v)}{\partial z} + \frac{\partial (P A_f)}{\partial t} + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) \quad (16)$$

If the value of enthalpy h is known from the time evolution of the energy equation, and pressure P is known from

measurement, then the temperature T can be reconstructed based on the departure function. The departure function is a mathematical function that characterizes the deviation of a thermodynamic property (enthalpy, entropy, and internal energy) of a real substance from that of an ideal gas at the same temperature and pressure. As presented by Gmehling et al. [28], for the Peng-Robinson equation of state, the enthalpy departure function is defined by Equation 17.

$$h - h^{id} = RT \left[T_r(Z - 1) - 2.078(1 + \kappa) \sqrt{\alpha(T)} \ln \left(\frac{Z + (1 + \sqrt{2})B}{Z + (1 - \sqrt{2})B} \right) \right] \quad (17)$$

where α is defined as $(1 + \kappa(1 - \sqrt{T_r}))^2$, T_r is the reduced temperature, P_r is the reduced pressure, Z is the compressibility factor, κ is a quadratic function of the acentric factor and B is calculated as $0.07780 \frac{P_r}{T_r}$.

Equation 17 requires an reference state, which is assumed to be $T_{ref} = 298.15$ K and $P_{ref} = 1.01325$ bar.

A root-finder can be used to find a value of temperature, which minimizes the difference between the value of enthalpy coming from the heat balance and the departure functions. The root finding procedure to repeated at every time step to find a temperature profile along spatial direction z .

$$\min_T \left(\underbrace{h(t, x)}_{\text{Heat balance}} - \underbrace{h(T, P, \rho_f(T, P))}_{\text{Departure function}} \right)^2 \quad (18)$$

2.2.8. Pressure term

As explained in Chapters 2.1, this low velocity region, pressure is, nearly constant due to the small pressure wave propagation that occurs at the speed of sound. Under such conditions, the term $\partial P / \partial t$ can be approximated by a difference equation, describing the pressure change in the whole system. The pressure P in the system is considered a state variable, while the pressure in the new time-step P_{in} is considered a control variable.

$$\frac{\partial P}{\partial t} \approx \frac{P_{in} - P}{\Delta t} \quad (19)$$

Such a simplified equation allows for instantaneous pressure change in the system but does not consider a gradual pressure build-up and the effects of pressure losses. In a real system, the dynamics of pressure change would depend on a pump and a back-pressure regulator.

2.2.9. Extraction yield

The process yield is calculated according to Equation 20 as presented by Sovova et al. [16]. The measurement equation evaluates the solute's mass at the extraction unit outlet and sums it up. The integral form of the measurement (Equation 20) can be transformed into the differential form (Equation 21) and augmented with the process model.

$$y = \int_{t_0}^{t_f} \frac{F}{\rho_f c_f} \Big|_{z=L} dt \quad (20)$$

$$\frac{dy}{dt} = \frac{F}{\rho_f c_f} \Big|_{z=L} \quad (21)$$

2.2.10. Initial and boundary conditions

It is assumed that the solvent is free of solute at the beginning of the process $c_{f0} = 0$, that all the solid particles have the same initial solute content c_{s0} , and that the system is isothermal, hence the initial state is h_0 . The fluid at the inlet is considered not to contain any solute. The initial and boundary conditions are defined as follows:

$$\begin{aligned} c_f(t=0, z) &= 0 & c_s(t=0, z) &= c_{s0} & h(t=0, z) &= h_0 \\ c_f(t, z=0) &= 0 & h(t, z=0) &= h_{in} & \frac{\partial c_f(t, z=L)}{\partial x} &= 0 \\ \frac{\partial h(t, z=L)}{\partial x} &= 0 & c_s(t, z=\{0, L\}) &= 0 & y(0) &= 0 & P(0) &= P_0 \end{aligned}$$

2.2.11. Discretization methods

The method of lines is used to transform the process model equations into a set of ODEs denoted as $G(x; \Theta)$. The backward finite difference is used to approximate the first-order derivative, while the central difference scheme approximates the second-order derivative z direction. The length of the fixed bed is divided into N_z , i.e. equally distributed points in the z direction. The state-space model after discretization is denoted as x and defined as follows:

$$\dot{x} = \frac{dx}{dt} = \begin{bmatrix} \frac{dc_{f,1}}{dt} \\ \vdots \\ \frac{dc_{f,N_z}}{dt} \\ \frac{dc_{s,1}}{dt} \\ \vdots \\ \frac{dc_{s,N_z}}{dt} \\ \frac{dh_1}{dt} \\ \vdots \\ \frac{dh_{N_z}}{dt} \\ \frac{dP}{dt} \\ \frac{dy}{dt} \end{bmatrix} = \begin{bmatrix} G_1(c_f, c_s, h; \Theta) \\ \vdots \\ G_{N_z}(c_f, c_s, h; \Theta) \\ G_{N_z+1}(c_f, c_s, h; \Theta) \\ \vdots \\ G_{2N_z}(c_f, c_s, h; \Theta) \\ G_{2N_z+1}(c_f, c_s, h; \Theta) \\ \vdots \\ G_{3N_z}(c_f, c_s, h; \Theta) \\ G_{3N_z+1}(c_f, c_s, h; \Theta) \\ \vdots \\ G_{3N_z+2}(c_f, c_s, h; \Theta) \end{bmatrix} = \underbrace{\begin{bmatrix} G_1(c_f, c_s, h; \Theta) \\ \vdots \\ G_{N_z}(c_f, c_s, h; \Theta) \\ G_{N_z+1}(c_f, c_s, h; \Theta) \\ \vdots \\ G_{2N_z}(c_f, c_s, h; \Theta) \\ G_{2N_z+1}(c_f, c_s, h; \Theta) \\ \vdots \\ G_{3N_z}(c_f, c_s, h; \Theta) \\ G_{3N_z+1}(c_f, c_s, h; \Theta) \\ \vdots \\ G_{3N_z+2}(c_f, c_s, h; \Theta) \end{bmatrix}}_{G(x; \Theta)}$$

where $x \in \mathbb{R}^{N_x=3N_z+2}$ and $\Theta \in \mathbb{R}^{N_\Theta=N_\theta+N_u}$, N_θ is the number of parameters, N_u is the number of control variables.

For a derivative to be conservative, it must form a telescoping series. In other words, only the boundary terms should remain after adding all terms coming from the discretization over a grid, and the artificial interior points should be cancelled out. Discretization is applied to the conservative form of the model to ensure mass conservation.

3. Optimal design of experiment for model discrimination

Mathematical models are constructed based on specific hypotheses regarding physical phenomena and are subsequently validated using experimental data. Each model predicts the response y as a function of experimental conditions Ξ and model parameters θ (parameters associated with extraction kinetics) and constitutes a subset of the parameter space Θ . Although the models differ in their structures and parameter sets, certain parameters may retain consistent physical interpretations across models.

The Appendix ?? introduces an alternative formulation to the model described in Section ??, specifically replacing the extraction kinetic term from Section 2.2.4. When multiple models are available, selecting the final model is a critical decision. A straightforward approach is to select the model with the lowest modelling error; however, this method is inadequate when several models exhibit similar performance. Alternatively, the relative evidence supporting one model over another can be evaluated. If such comparisons remain inconclusive, a new experiment should be designed at the operating conditions where the model outputs differ most significantly.

The single-point design of experiment for model discrimination, as presented in Section ??, can be extended to accommodate multiple data points. In this approach, the system is operated such that the model outputs exhibit maximal statistical divergence, based on information obtained from prior experiments. The criterion for selecting subsequent experiments incorporates posterior probabilities as weighting factors. Following Himmelblau [30], the objective function is defined as:

$$K_v = \frac{1}{2} P_1^{(n)} P_2^{(n)} \cdot j_{DJ} \quad (22)$$

Here, j_{DJ} denotes the Jeffreys divergence between the predictive distributions of the two models. The term $P_1^{(n)} P_2^{(n)}$ weights the design according to the current posterior model probabilities, as described in Section ???. When one model has been effectively eliminated, K_v remains small under all experimental conditions.

Both models assume normally distributed errors, with predicted covariance matrices Σ_1 and Σ_2 , respectively (see Appendix ??):

$$p_i(y|\Xi) = \frac{1}{\sqrt{(2\pi)^{n_y} |\Sigma_i|}} \exp\left(-\frac{1}{2}(Y(\Xi) - y_i(\Xi))^T \Sigma_i^{-1} (Y(\Xi) - y_i(\Xi))\right) \quad (23)$$

Here, $Y(\Xi)$ denotes the observed data, $y_i(\Xi)$ represents the predicted response from model i , and Ξ indicates the operating conditions. The model parameters are fixed at $\hat{\theta}$, so $p_i(y|\Xi, \hat{\theta})$ is abbreviated as $p_i(y|\Xi)$.

Prior to conducting a subsequent experiment, the output $Y(\Xi)$ remains unknown, which precludes direct computation of D_{KL} . Assuming Model 1 is correct and the observations are normally distributed, a closed-form expression for D_{KL} divergence can be derived as follows:

$$D_{\text{KL}}(p_1(y|\Xi)\|p_2(y|\Xi)) = \frac{1}{2} \left[\ln \left(\frac{|\Sigma_2|}{|\Sigma_1|} \right) - n_Y + \text{tr}(\Sigma_2^{-1}\Sigma_1) + (y_2(\Xi) - y_1(\Xi))^T \Sigma_2^{-1} (y_2(\Xi) - y_1(\Xi)) \right] \quad (24)$$

Analogously, $D_{\text{KL}}(p_2(y|\Xi)\|p_1(y|\Xi))$ can be obtained by interchanging the indices. Because the correct model is unknown, both quantities are combined to define the objective function j_{DJ} . This symmetric combination of two KL divergences is referred to as the Jeffreys divergence:

$$j_{DJ} = D_{\text{KL}}(p_1(y|\Xi)\|p_2(y|\Xi)) + D_{\text{KL}}(p_2(y|\Xi)\|p_1(y|\Xi)) = \int_{-\infty}^{\infty} [p_1(y|\Xi) - p_2(y|\Xi)] \ln \left(\frac{p_1(y|\Xi)}{p_2(y|\Xi)} \right) dy \quad (25)$$

A large value of j_{DJ} is obtained when $p_2(y|\Xi)$ differs substantially from $p_1(y|\Xi)$, or vice versa. In either case, the outcome indicates a strong preference for one model over the other. By expanding the objective function, its algebraic form can be obtained:

$$j_{DJ} = \frac{1}{2} [\text{tr}(\Sigma_1^{-1}\Sigma_2 + \Sigma_2^{-1}\Sigma_1) - 2n_Y + (y_2(\Xi) - y_1(\Xi))^T (\Sigma_1^{-1} + \Sigma_2^{-1}) (y_2(\Xi) - y_1(\Xi))] \quad (26)$$

Maximising j_{DJ} encourages operating conditions in which the predictive distributions of competing models are maximally separated, given the associated uncertainties. Experimental data generated under the true model are expected to yield a lower likelihood when assessed under the alternative hypothesis. Owing to the symmetry of the Jeffreys divergence, the solution to the optimisation problem suggests that one of the two hypotheses is correct. This method assumes that one of the models will accurately predict the data from the subsequent experiment.

3.0.1. Problem formulation

The optimisation problem (Equation 27) is addressed by varying temperature, mass flow rate, and pressure. Data are assumed to be collected cumulatively, adhering to the sampling intervals established in the initial experiments. Decision variables are normalised between -1 and 1, with penalty weights set to 0.001. The inlet temperature and mass flow rate are adjusted every 10 min, whereas the pressure is modified every 30 min. The parameter τ_P is set to 120 sec to balance system dynamics and the smoothness of pressure derivatives. Notably, system pressure reaches the step-change value after 10 min, resulting in the overlap between piece-wise control intervals and system pressure for most of the experiment.

$$\begin{aligned} \Xi^* &= \arg \min_{T^{in}(t), F(t), P(t)} -(j_{DJ} + j_{\text{smooth}}) \\ \text{subject to } \dot{x}(t) &= G(x(t), t, \Theta; \Xi), \\ t_0 &= 0 \text{ min}, \\ t_f &= 600 \text{ min}, \\ 30^\circ\text{C} &\leq T^{in}(t) \leq 40^\circ\text{C}, \\ 3.3 \times 10^{-5} &\leq F(t) \leq 6.7 \times 10^{-5}, \\ 100 &\leq P(t) \leq 200 \text{ bar}, \\ j_{\text{smooth}} &= \alpha_T \sum_{k=1}^{N_T-1} (T_{k+1}^{in} - T_k^{in})^2 + \alpha_F \sum_{k=1}^{N_F-1} (F_{k+1} - F_k)^2 + \alpha_P \sum_{k=1}^{N_P-1} (P_{k+1} - P_k)^2 \end{aligned} \quad (27)$$

	No modelling error	Empirical σ
AUC	1.0	0.89
KS	1.0	0.62
D_J	0.69	0.27

Table 1

Single-point discrimination diagnostics of the optimal trajectory

3.0.2. Numerical results

Figure 5b compares the initial and final values of the cost function. Most results reached an objective function value of 45.5, indicating convergence to a global solution. The optimal control profiles (Figures 5d–5f) exhibit a two-phase structure. During the first phase, from 0 to 200 min, the inlet temperature remains nearly constant at their initial values, while the pressure rises. In the second phase (200 min onward), the inlet temperature increases gradually, while the pressure stays at the same level. The mass flow rate is kept at its upper bound during the entire batch.

Figure 5g shows that, initially, the predicted yield curves from both models follow similar trajectories but diverge over time. The Jeffreys divergence objective maximises the statistical separation of the output distributions over entire trajectories, rather than pointwise differences. Analysis of the output distributions at the final time reveals an overlap (Figure 5e), which warrants further investigation. Table 1 presents the discrimination measures evaluated at the final time. In the absence of measurement error, the optimised trajectory achieves perfect separation (AUC = 1.0, KS = 1.0). When empirical measurement noise is introduced, the AUC decreases to 0.89 and the Jensen–Shannon divergence to 0.27. These values indicate only a modest improvement over the best constant-condition scenario (Figure ??). The screening maps in Figures ?? and ?? clarify this result: the high-discrimination region at elevated pressure and flow rate is broad and relatively flat, so constant conditions already capture most of the discrimination capability.

Interpretation of the results should consider that the objective function presumes one of the two models accurately represents the data from the subsequent experiment. If both models are inadequate, the designed experiment may only achieve high discrimination between two incorrect alternatives. In such a scenario, a subsequent experiment conducted according to the derived procedure Ξ^* would not correspond to any of the predicted yield curves. This outcome would indicate insufficient modelling assumptions and advocate for the model redevelopment.

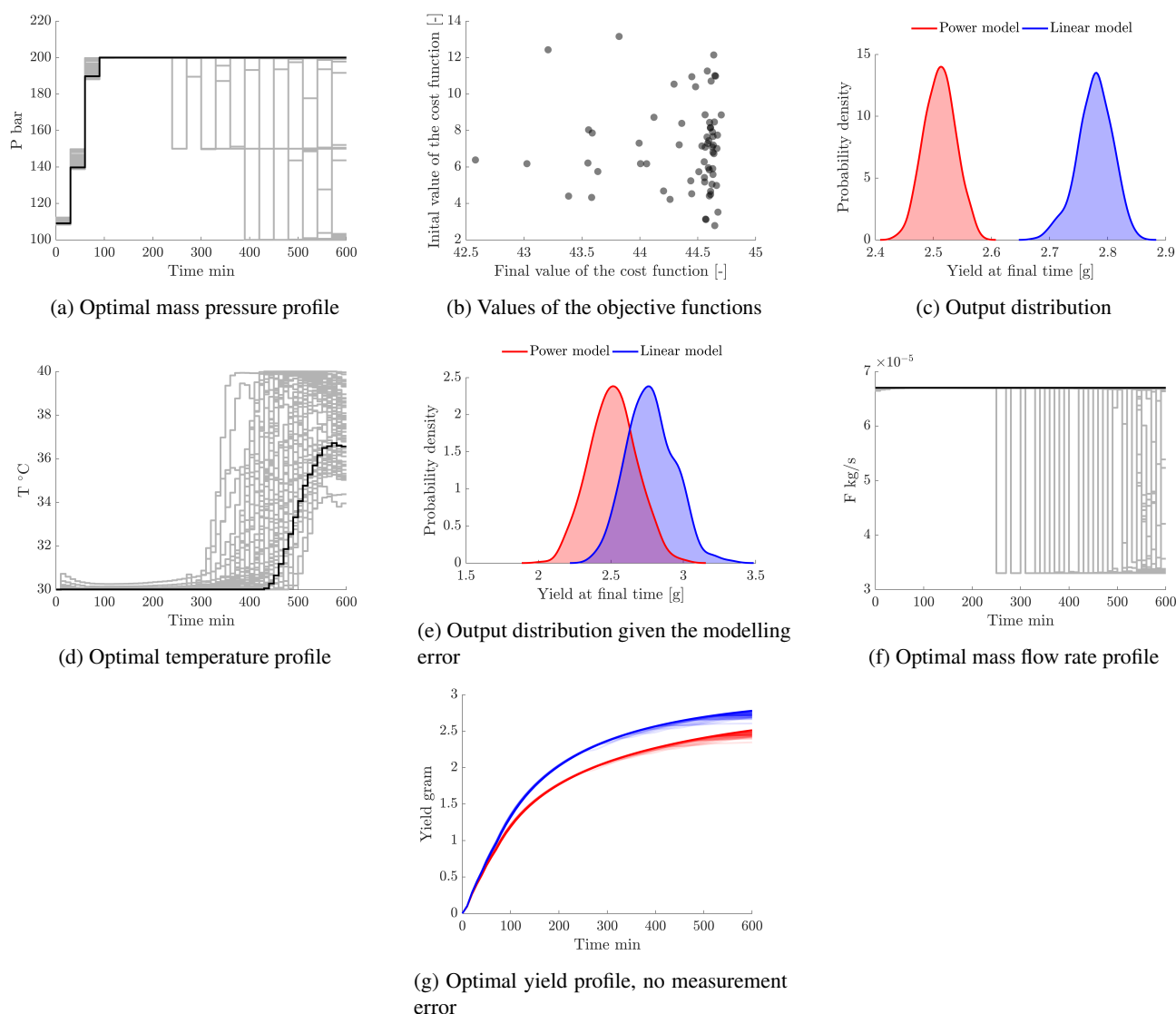


Figure 5: Optimal profiles for the model discrimination

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